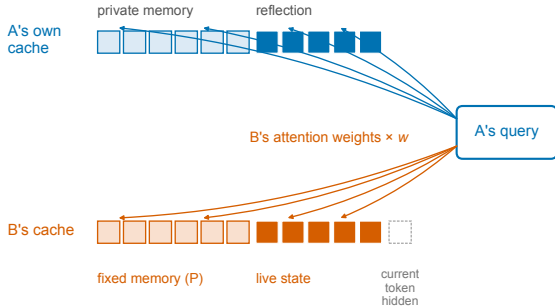


## a A's attention at every layer



## b What A may read

### fixed memory only (KV-P)

B's private prompt and note; main setting  $w = 2$  (one-way or two-way)

### memory + live state (KV-ALL)

also B's ongoing reflection; with two-way reading, a feedback loop

### separate weights (KV-PC)

memory at  $w_P$ , live state at  $w_C$ ;  $w_C \rightarrow 0$  recovers KV-P

Output:  $o = o_{own} + \beta(o_B - o_{own})$ , with  $\beta = \sigma(\ell_B + \log w - \ell_{own})$ , where  $\ell$  are log-sum-exp attention scores.  $w = 0$  gives  $\beta = 0$  exactly.